LLM Reasoning Paper Outline

Three Routes to Better LLM Reasoning: Preference Learning, Verifiable Rewards, and Test-Time Compute

Topic 6 | Mentor Kaiwei Lei

Research question: How do preference learning, verifiable rewards and additional inference-time computation improve reasoning?

Table of Contents

Abstract
Summarize the research question, literature review approach, and main findings.

1. Introduction
Introduce the motivation, research question, and scope of the comparison.

2. Background and Related Work
Explain RLHF, verifiable rewards, and test-time compute, and position the paper within the relevant literature.

3. Methodology
Describe how the literature is selected, analyzed, and compared using consistent criteria.

4. Three Routes to Improved Reasoning
Examine how preference learning, verifiable rewards, and additional inference computation can improve reasoning.

- **4.1 Preference Learning**
  Explain RLHF and DPO and discuss their reported benefits, costs, and limitations.
- **4.2 Verifiable Rewards and DeepSeek-R1**
  Describe RLVR and the DeepSeek-R1 training pipeline, including the main evidence and limitations.
- **4.3 Test-Time Compute**
  Explain candidate generation, verification, revision, and adaptive compute allocation using the study by Snell and colleagues.

5. Comparison and Discussion
Compare the approaches in terms of reasoning performance, evaluation metrics, compute costs, and failure modes.

6. Conclusion and Future Work
Answer the research question and identify the main open questions for future research.

References
List the verified sources in IEEE citation style.